

6. Digestion of food is required to provide nutrition for the human body.

(a) (i) Explain how digestion provides nutrition.

[2]

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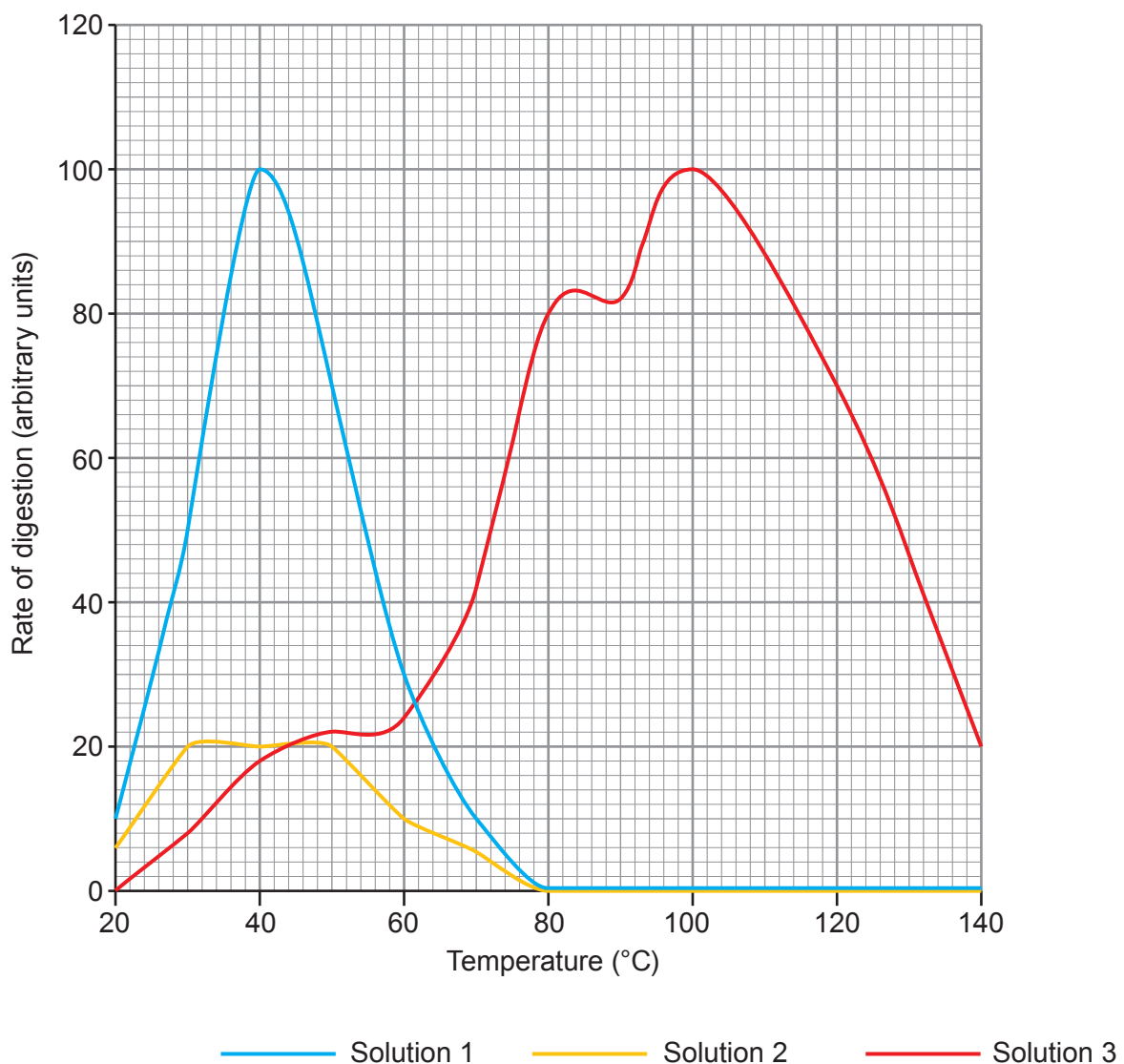
(ii) Describe the process of peristalsis in the digestive system.

[2]

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(b) A group of students were given two solutions of a human enzyme (**solution 1 and solution 2**) and one solution of bacterial enzyme (**solution 3**). All solutions were the same concentration and volume. They measured the rate of digestion using each enzyme solution. A computer calculated the rate of digestion and the results are shown below.



Use the data in the graph and your knowledge to answer the following questions.

- (i) Using the results for **solution 1**, describe and explain the effect of temperature on the rate of digestion between 20 and 60 °C. [4]

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- (ii) Suggest **one** reason why the results shown for **solution 2** are different to **solution 1**. [1]

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- (iii) State a suitable control in this experiment. [1]

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- (iv) **Solution 3** contains an enzyme from a species of bacteria that is found in a hot volcanic region.

Describe the differences between the action of this bacterial enzyme and the enzyme found in humans. [2]

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